

Asjad Nirban

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EDUCATION

University of Virginia Charlottesville, VA, United States	<i>Expected Dec 2025</i>
<i>Master of Science in Computer Science</i>	<i>GPA: 3.76/ 4.00</i>
University of Mumbai Mumbai, India	<i>May 2023</i>
<i>Bachelor of Engineering in Computer Engineering</i>	<i>GPA: 3.73/ 4.00</i>

PROFESSIONAL EXPERIENCE

Data Engineering Research Assistant <i>Darden School of Business, University of Virginia, USA</i>	<i>Mar 2025 – Present</i>
<i>Tech Stack: Python SQL HPC Pandas AWS Selenium</i>	
- Extracted and processed large-scale data from EORA trade tables, SEC mutual fund filings, and legal records, building matrices with 15,000+ columns and millions of rows using Pandas, Polars, SQL and IP hopping techniques - Engineered automated ETL pipelines and classification systems, reducing data processing time by 60% and accelerating research across UVA’s finance, law, and economics departments - Deployed large-scale data extraction and transformation scripts on High-Performance Computing (HPC) servers and AWS, enabling fast, scalable processing of complex datasets, websites, and multi-format documents	
Software Developer Intern <i>Crayon Infotech Pvt Ltd., India</i>	<i>Jan 2024 – Mar 2024</i>
<i>Tech Stack: CodeIgniter Laravel Python Git Algorithms C/C++ SQL Flask API Data Structures</i>	
- Built 5 full-stack apps with Laravel and CodeIgniter, ensuring 99.9% uptime and resolving 100% of assigned bug tickets - Designed and optimized SQL queries and schemas for 1M+ record databases, reducing query time by ~40% under high traffic. Streamlined data access and improved backend efficiency in modules handling large transactional workloads - Created and tested RESTful APIs for seamless integration with internal services and third-party tools. Collaborated with QA to debug critical issues and accelerate stable deployment cycles	
Blockchain Research Intern <i>Lab Systems Pvt. Ltd., India</i>	<i>Jul 2022 – Feb 2023</i>
<i>Tech Stack: Blockchain Architecture Blockchain Node Hashing Solidity Smart Contracts Ganache IPFS</i>	
- Set up and configured 3+ cryptocurrency nodes, including Bitcoin, to support fraud analysis and data exploration - Developed Python scripts and OSINT workflows to investigate suspicious transactions and support forensic research - Trained and led a 12-member team in blockchain investigation methods, enhancing technical capabilities and tool usage	

PROJECTS

Synthetic Medical Image Generation <i>Tech Stack: GANs, Python</i>
- Trained and optimized GAN models for generating synthetic blood cancer cell datasets, enhancing data diversity - Generated synthetic images with 85% similarity with real data, demonstrating high perceptual quality and effectiveness
Visual Room Interfaces <i>Tech Stack: Stable Diffusion 3.5 Large Turbo Model, GenAI, Python, Influxdb, Sensors</i>
- Designed a system using Stable Diffusion to create real-time visualization of environmental conditions using sensor data - Streamlined process of interpreting raw data for improving decision-making process, achieving 90% success in user study
Automated Business Card Processor <i>Tech Stack: Python, Tesseract OCR, Flask, Pandas, SQLite, WhatsApp API, JSON</i>
- Engineered a machine learning model with integrated OCR libraries for fully automated business card data extraction - Automated processing and extraction of details from business cards, outperforming 94% of available paid APIs
Medilog: A Blockchain-Based Pharmaceutical Supply Chain System <i>Tech Stack: Ethereum, Solidity, MERN stack</i>
- Architected and deployed an end-to-end pharmaceutical supply chain solution leveraging blockchain technology - Enabled 100% tamper-proof tracking of medical drugs and presented solution to the Indian Medical Association
Autonomous Driving Fault Simulation Platform <i>OpenPilot, MetaDrive, ScenarioNet, Python, Linux</i>
- Integrated Waymo dataset into MetaDrive using ScenarioNet to simulate real-world scenarios in lightweight environments - Analyzed OpenPilot’s behavior across 3000+ scenarios, revealing critical limitations in pedestrian and cyclist detection - Created a Python-based filtering pipeline that excluded 50% unsupported scenarios, enhancing OpenPilot accuracy

PATENT & PUBLICATIONS

MediLog– (Granted), German Patent File No - DE202023102823U1, patent-link Research Paper: "Face Mask Detection and Attendance System" IRJET (Impact Factor: 7.529) paper-link Research Paper: "Crowdfunding for Engineering Startups Using Blockchain" IJRAR (Impact Factor: 7.17) paper-link

SKILLS AND CERTIFICATIONS

Programming Languages: Python, C++/C, Java, JavaScript, SQL, NoSQL, PHP, HTML, CSS Frameworks & Libraries (Software): ReactJS, NextJS, NodeJS, CodeIgniter, Laravel, SQLite, Bootstrap, Flask Frameworks & Libraries (AI/ML): TensorFlow, PyTorch, Keras, Pandas, Selenium, NumPy, Scikit-learn, GenAI Tools & Platforms: AWS, MongoDB, MySQL, Flask API, Git/GitHub, VS code, JIRA Technical Skills: Data Structures & Algorithms, OOPs, ETL, Data Analytics, AI/ML, NLP, OS, Distributed Systems Certifications: AWS Cloud Foundations, CCNav7 (Networks, Switching & Routing, Enterprise Networking)
